Alicia (Sudlerd) Pankka

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Software Engineer with years of experience in games and full-stack development, specialized in API integration, UI, shaders, animation, and GIS development. I built and launched CosOverHere.com as a solo dev, from the Next.js frontend to the API, database, and deployment. I initially assisted in shipping features on Ragnarok: Battle Academy and built production. Experienced in leading and mentoring junior developers under tight deadlines, and in using AI and MCP in daily development.

Education

MSc in Computer Science (Distributed Systems), University of Tartu, Estonia (2022-2024)
BSc in Science and Technology (Game Art & Design), Thammasat University, Thailand (2015-2019)

Professional Experience

XR Developer| **Helsinki XR Center** (HXRC) (Jan 2025 - Present)

- Initiate and develop several RDI related to XR and geospatial data, digital twin projects.
- Organize development tasks, estimate difficulty, and distribute multiple simultaneous tasks.
- Built tool pipelines and transformation logic across projects to facilitate the workflows.
- Define technical architecture and contribute to a scalable, modular, and maintainable setup.
- Collaborate with teams and partner companies throughout development cycles.
- Lead and supervise juniors and team members in high-pressure project environments.
- Organize in-house showroom visits and workshops, after-work sessions, and larger events.

Frontend: Unity3D Developer | Wildchain (Digital Wild Animals) (Jan 2022 - Oct 2022)

- Develop an authentication system and cooperate with a backend team
- Design and create UI architecture and logic
- Display data from API to uGUI and in-game sequences using UniRx
- Cooperate with the team designer to visualise features correctly
- Resolve technical debt before release for senior developers
- Polish in-game graphics system

Unity3D Developer | Music Ed-tech, BNK Musicmall Co.,Ltd - Notero (June 2021 - Dec 2021)

- Connect the game and the MIDI Keyboard using the MIDI Unified plugin
- Initialise game logic and classes
- Maintain up-to-date algorithms based on UML from a senior developer
- Discuss with other developers how to link data between game-sequences
- Create responsive prefabs: UI, the visual piano and raindrops

Unity3D Developer | Ragnarok: Battle Academy, Electronic Extreme (Jan 2020 - May 2021)

- Manage and develop 3D asset pipelines and automation.
- Develop inventory and shop features through Unity Editor and frontend.
- Implement an authentication and experience with Firebase.
- Collaborate with UI artists to integrate UI into Unity.

Relevant Work Experience (Internships)

UE5 Integration Engineer (Flair), Artineering (Mar 2024 - May 2024)

- Prototype the new version in C++ UE5, focusing on making stylised HLSL shaders.
- Work closely with the team lead to adapt the original Flair tools to be compatible with UE5.
- Implement documentation to assist team members in understanding the current architecture and utilizing the new tools effectively.

C++ and Linux Administration (LizardFs, SaunaFS), Leil Storage (Diaway) (July 2023 - Aug 2023)

- Develop Dockerfiles to containerize LizardFS services for development and test environments.
- Automate the Debian build packages pipeline and deployment workflows using Bash scripting.
- Engaging in the DevOps process, working with tools like Docker and integration tests.
- Contribute to client-server interactions by working with RPC and user session management.
- Manage user sessions on the admin side and communicate with the server using LizardFs RPC.
- Explore its chunk server and modified source code within a distributed filesystem codebase.

Professional Projects

CosOverHere.com — Full-Stack Cosplay GIS Map (Nordic/Baltic)

Frontend (Client)

- Build an interactive GIS web app that maps anime conventions and cosplay-friendly venues.
- Render a 3D vector map with MapLibre GL.
- Filter venues by city, click to fly to a location, and zoom in for detail cards.
- Implement a navigable event calendar, pagination, and multi-platform support
- Implement an admin dashboard for creating, editing, and deleting events.
- Design the app to run standalone with sample data, then wire it to the backend.

Backend (Server)

- Architect as a monorepo of deployable Cloudflare Workers microservices.
- Build REST APIs with Hono, backed by a SQLite (D1), covering full CRUD.
- Defined a shared, schema-validated data contract (with Zod) to keep services consistent.
- Create a proxy service that gets and caches official geometry from external mapping services.

AAXLP VR Factory Simulation

- Implement a factory system in Unity with a runtime workstation and tool creation
- Design a data export model for user state tracking and persistence (JSON-based)
- Initiate a structured architecture following the SOLID principle.
- Lead and supervise junior contributors under high-pressure development conditions
- Collaborate with the client during optimization and testing phases
- Integrate the Unity VR-based project into the cloud platform (VIROO)

ENPA (Kaupunkidata – City Data Integration)

- Integrate geospatial datasets into XR digital twin systems using API-driven workflows.
- Work with Mapbox SDK for spatial visualization and interaction in WebGL environments.
- Develop data pipelines for aggregating environmental and energy datasets
- Build and develop data pipelines for aggregating energy data (daily, weekly, monthly).

VLL (Virtual Learning Platform)

- Develop **full-stack** features across backend (NestJS), client applications (student/teacher), and shared in-house SDK systems.
- Implement classroom management systems and APIs supporting interactions.
- Integrate multiplayer functionality using Photon Fusion for live classroom scenarios.
- Build FSM NPC systems and integrate with the Convai conversation tool, connecting dialogue with state machines and contextual behavior.

Other projects: KebabLala, WaterSort with shaders, FlairUE5

CORE SKILLS

Game Development

- C#, Unity (UI Toolkit,uGUI, Editor)
- HLSL/GLSL, Photon Fusion, Firebase
- Mobile game development (Android)
- Addressable assets
- Dotween, Convai, Animation pipelines

Backend

- Python, Flask, NestJs, OpenAPI
- REST API design, async services
- Data processing, ETL logic

Mobile / Platform (Familiarity)

- Android fundamentals
- Mobile SDK concepts
- E2E test

Databases

- Relational DB: PostgreSQL and MySQL

Frontend

- React / TypeScript
- Next.js / Node.js
- Component-based UI architecture, state management, API-driven UIs

DevOps / Cloud

- Docker, Docker Compose, Kubernetes, Linux environments(Ubuntu)
- CI/CD: GitHub Actions, GitLab CI
- Basic AWS knowledge (S3, EC2, Lambda concepts)

AI / LLM EXPERIENCE

- Integrated LLM APIs (OpenAI, Claude, Deepseek) into applications for prompt automation, dialogue systems, and contextual interactions.
- Worked with conversational AI systems (Convai, Claude Opus 3) in Unity, connecting NPC behavior, state transitions, and real-time interactions to AI-generated responses.

CERTIFICATIONS

- TESOL Certification (communication, teaching, mentoring)
- AWS & DevOps self-study coursework (containers, CI/CD, monitoring)

LANGUAGES

- English (C2 – fluent)
- Finnish (A2 – basic working)